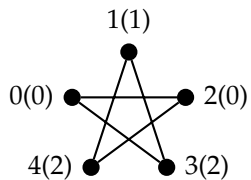
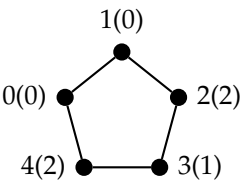


Graph Isomorphism Problem

Two undirected simple graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$, with $|V_G| = |V_H|$ and $|E_G| = |E_H|$, are isomorphic, if there exists a bijective mapping $f : V_G \rightarrow V_H$ such that:

$$\forall u, v \in V_G : (u, v) \in E_G \Leftrightarrow (f(u), f(v)) \in E_H$$

Apart from checking connectivity, you also need to check if the attributes of the two vertices are the same for the mapped vertices. e.g. In the graphs G and H below, the vertices are number with an attribute. In G , vertex 0 has an attribute 0, and 4 has an attribute 2. The two graphs are isomorphic with the mapping f .

G	H	f
		$\begin{aligned} 0 &\rightarrow 0 \\ 1 &\rightarrow 3 \\ 2 &\rightarrow 1 \\ 3 &\rightarrow 4 \\ 4 &\rightarrow 2 \end{aligned}$

- Write a function that returns a mapping or None if two graphs G and H are either isomorphic or not.

```
def is_isomorphic(G, V):
    mapping = None
    return mapping
```

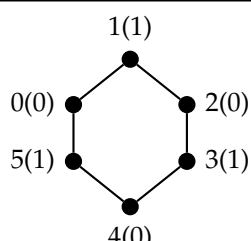
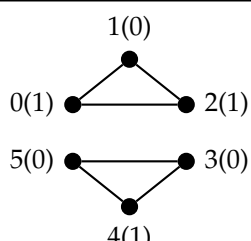
- The graphs G and H are input as a pair of elements ([list of vertex attributes], [list of edges, where an edge is a pair of vertices]). Vertices are labeled as numbers between 0 and $N-1$, where N is the size of attribute list. For example the above two graphs will be input as:

```
G = ([0, 1, 0, 2, 2], [(0, 2), (0, 3), (1, 3), (1, 4), (2, 4)])
H = ([0, 0, 2, 1, 2], [(0, 1), (1, 2), (2, 3), (3, 4), (4, 0)])
mapping = is_isomorphic(G, V)
```

- Return value is a list of N numbers 0 and $N-1$ where the index of the number is the vertex in G and the value is the vertex in H . For example, the mapping in the figure above will be output as:

```
print(mapping)
# should print one solution such as [0, 3, 1, 4, 2]
```

- Example where two graphs are not isomorphic:

G	H
	

```
G = ([0, 1, 0, 1, 0, 1], [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 0)])
H = ([1, 0, 1, 0, 1, 0], [(0, 1), (1, 2), (2, 0), (3, 4), (4, 5), (5, 3)])
mapping = is_isomorphic(G, H)
print(mapping)
# should print None
```